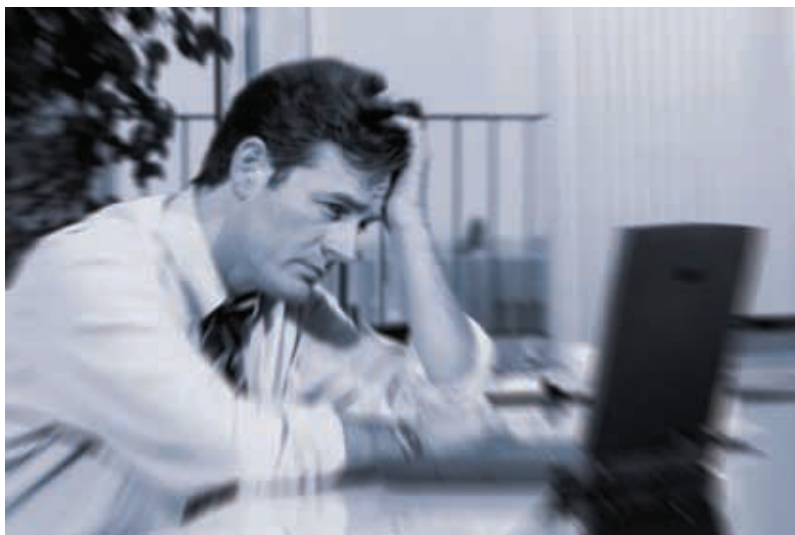




highly depend on the resources one has. Even the latest Intel Core i7 quad core or AMD 12-core processor won't be able to get to a password beyond a certain complexity.

So how much complexity are we talking? What resources? And what are the reasonable limits?

When we talk about complexity, calculating it depends on what all characters are permissible in a password. Let's say we have a password which could be made up of the 26 letters of the alphabet and their capital counterparts, 10 digits, and 32 special characters on a standard keyboard (~ ` ! @ # \$ % ^ & * () - _ = + \ | [{] } ; : ' " , < . > / ?) we have 94 characters, plus the white-space character makes it 95. This means for a single digit password we have 95 possibilities. For a simple two character password, we have as many as $95 \times 95 (= 9,025)$ combinations. For a password of 6 characters, we have $95 \times 95 \times 95 \times 95 \times 95 \times 95 (= 735,091,890,625)$ combinations. For any arbitrary n-character password we can see that there are 95^n combinations. It is important to note that a password cracker will need to test all combinations starting from a lowly single character – if such passwords are permissible – to increasingly long combinations till the password is found. So for password up to n characters, and with a minimum password length of m, the attacker will need to perform $95^m + 95^{m+1} + 95^{m+2} + \dots + 95^n (= \sum_{i=m}^n 95^i)$ tests.

Well, look at it this way; if you have a computer which is capable of testing a million passwords each second, a 6 character password could take as much